

三. (15 分) 如图所示, 两块无限大理想导体板, 相距为 a , 若已知其中的电场 $E_z = E_x = 0$,

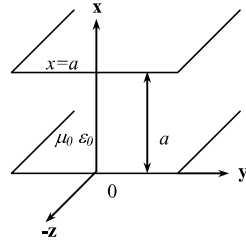
而满足麦克斯韦方程的 E_y 为: $E_y = E_0 [\sin k_x x + A \cos k_x x] e^{-jk_z z}$.

(1) 利用边界条件决定常数 A 和 x 方向传播常数 k_x , 写出 E_y 表达式;

(2) 求磁场强度表达式 $\mathbf{H}$;

(3) x 方向和 z 方向的平均坡印亭功率流;

(4) 两块导体板内壁上表面电流密度的表达式。



(1)解: 理想导体表面电场强度切线分量为零, 有:

$x=0$ 处, $E_y = 0$, 故必须 $A=0$

$x=a$ 处, $E_y = 0$, 故必须 $\sin k_x a = 0$

即: $k_x a = m\pi$, $k_x = \frac{m\pi}{a}$, $m=1, 2, \dots$

所以: $E_y = E_0 \sin\left(\frac{m\pi}{a}x\right) e^{-jk_z z}$ $m=1, 2, \dots$

(2)解: 根据麦克斯韦第二方程: $\nabla \times \mathbf{E} = -j\omega \mathbf{B} = -j\omega \mu_0 \mathbf{H}$

$$\nabla \times \mathbf{E} = \left(-\frac{\partial E_y}{\partial z} \mathbf{x}_0 + \frac{\partial E_y}{\partial x} \mathbf{z}_0\right) = -j\omega \mu_0 \mathbf{H}$$

将 E_y 代入得:

$$jk_z E_0 \sin\left(\frac{m\pi}{a}x\right) e^{-jk_z z} \mathbf{x}_0 + \frac{m\pi}{a} E_0 \cos\left(\frac{m\pi}{a}x\right) e^{-jk_z z} \mathbf{z}_0 = -j\omega \mu_0 \mathbf{H}$$

磁场强度 $\mathbf{H}$ 有两个分量:

$$H_x = -\frac{k_z E_0}{\omega \mu_0} \sin\left(\frac{m\pi}{a}x\right) e^{-jk_z z}$$

$$H_z = \frac{j}{\omega \mu_0} \frac{m\pi}{a} E_0 \cos\left(\frac{m\pi}{a}x\right) e^{-jk_z z}$$

(3)解:

x 方向: $S_x = \frac{1}{2} \text{Re}(\mathbf{E}_y \times \mathbf{H}_x^*) = \frac{1}{2} \text{Re}\left[E_0 \sin\left(\frac{m\pi}{a}x\right) j \frac{1}{\omega \mu_0} \frac{m\pi}{a} E_0 \cos\left(\frac{m\pi}{a}x\right)\right] = 0$

z 方向: $S_z = -\frac{1}{2} \text{Re}(\mathbf{E}_y \times \mathbf{H}_z^*) = \frac{1}{2} \text{Re}\left[E_0 \sin\left(\frac{m\pi}{a}x\right) \frac{k_z E_0}{\omega \mu_0} \sin\left(\frac{m\pi}{a}x\right)\right] = \frac{k_z E_0^2}{2\omega \mu_0} \sin^2\left(\frac{m\pi}{a}x\right)$

(4)解: 根据 $\mathbf{J}_s = \mathbf{n} \times \mathbf{H}$ 求内壁上的电流分布:

$$\mathbf{J}_s|_{x=0} = -\mathbf{y}_0 H_z|_{x=0} = -\mathbf{y}_0 \frac{j}{\omega \mu_0} \frac{m\pi}{a} E_0 e^{-jk_z z}$$

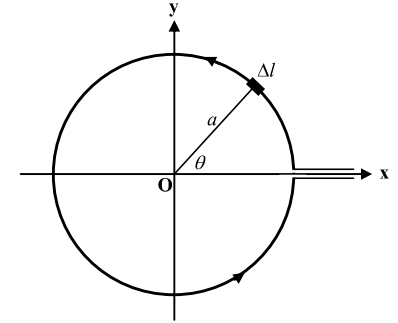
$$\mathbf{J}_s|_{x=a} = \mathbf{y}_0 H_z|_{x=a} = \mathbf{y}_0 \frac{j}{\omega \mu_0} \frac{m\pi}{a} E_0 \cos(m\pi) e^{-jk_z z}$$

五. (15 分) 如图所示, 一环形天线, 半径 a ($a \gg \lambda$), 环线上电流为 $I(\theta) = I_0 \sin(ka\theta)$ 。图中 z 方向为和纸面垂直方向。

积分公式
$\int \sin \alpha \sin(\gamma \alpha) d\alpha = \frac{\gamma \sin \alpha \cos(\gamma \alpha) - \cos \alpha \sin(\gamma \alpha)}{1 - \gamma^2}$
$\int \cos \alpha \sin(\gamma \alpha) d\alpha = \frac{\gamma \cos \alpha \cos(\gamma \alpha) + \sin \alpha \sin(\gamma \alpha)}{1 - \gamma^2}$

$$\int_0^{2\pi} \sin \alpha \sin(\gamma \alpha) d\alpha = \frac{\sin(2\pi\gamma)}{\gamma^2 - 1}$$

$$\int_0^{2\pi} \cos \alpha \sin(\gamma \alpha) d\alpha = \frac{-\gamma \cos(2\pi\gamma) + \gamma}{\gamma^2 - 1}$$



(1) 取环线上小段 $\Delta l = a \Delta \theta$, $\Delta l \ll \lambda$, 写出其在圆心处的辐射磁场和电场 (写出 x 、 y 、 z 方向的分量);

(2) 求整个环形天线在圆心处的辐射磁场和电场 (写出 x 、 y 、 z 方向的分量)。

(1)解:

$$\mathbf{H} = \mathbf{z}_0 \frac{jkI \Delta l e^{-jka}}{4\pi a} = \mathbf{z}_0 \frac{jkI_0 \sin(ka\theta) \Delta \theta e^{-jka}}{4\pi}$$

$$\begin{aligned} \mathbf{E} &= \mathbf{x}_0 \sqrt{\frac{\mu}{\epsilon}} \frac{jkI \Delta l e^{-jka}}{4\pi a} \sin \theta - \mathbf{y}_0 \sqrt{\frac{\mu}{\epsilon}} \frac{jkI \Delta l e^{-jka}}{4\pi a} \cos \theta \\ &= \mathbf{x}_0 \sqrt{\frac{\mu}{\epsilon}} \frac{jkI_0 \sin(ka\theta) \Delta \theta e^{-jka}}{4\pi} \sin \theta - \mathbf{y}_0 \sqrt{\frac{\mu}{\epsilon}} \frac{jkI_0 \sin(ka\theta) \Delta \theta e^{-jka}}{4\pi} \cos \theta \\ &= (\mathbf{x}_0 \sin \theta - \mathbf{y}_0 \cos \theta) \sqrt{\frac{\mu}{\epsilon}} \frac{jkI_0 \sin(ka\theta) \Delta \theta e^{-jka}}{4\pi} \end{aligned}$$

(2)解:

$$\begin{aligned} \mathbf{H} &= \mathbf{z}_0 \int_0^{2\pi} \frac{jkI_0 \sin(ka\theta) e^{-jka}}{4\pi} d\theta = \mathbf{z}_0 \frac{jkI_0 e^{-jka}}{4\pi} \left[-\frac{\cos(2\pi ka) - 1}{ka} \right] \\ &= \mathbf{z}_0 \frac{jkI_0 e^{-jka}}{4\pi} \left[\frac{1 - \cos(2\pi ka)}{ka} \right] = \mathbf{z}_0 \frac{jI_0 [1 - \cos(2\pi ka)] e^{-jka}}{4\pi a} \end{aligned}$$

$$\begin{aligned} \mathbf{E} &= \int_0^{2\pi} (\mathbf{x}_0 \sin \theta - \mathbf{y}_0 \cos \theta) \sqrt{\frac{\mu}{\epsilon}} \frac{jkI_0 \sin(ka\theta) e^{-jka}}{4\pi} d\theta \\ &= \sqrt{\frac{\mu}{\epsilon}} \frac{jkI_0 e^{-jka}}{4\pi} \int_0^{2\pi} (\mathbf{x}_0 \sin \theta - \mathbf{y}_0 \cos \theta) \sin(ka\theta) d\theta \\ &= \sqrt{\frac{\mu}{\epsilon}} \frac{jkI_0 e^{-jka}}{4\pi} \left(\mathbf{x}_0 \frac{\sin(2\pi ka)}{k^2 a^2 - 1} - \mathbf{y}_0 \frac{ka - ka \cos(2\pi ka)}{k^2 a^2 - 1} \right) \end{aligned}$$